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Synthesis of a monolayer fullerene network

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Two-dimensional (2D) carbon materials, such as graphene, have attracted particular attention owing to the exceptional carrier transport characteristics that arise from the unique π-electron system in their conjugated carbon network structure[1–4]. To complement zero-bandgap graphene, material scientists have devoted considerable effort to identifying 2D carbon materials[5–8]. However, it is a challenge to prepare large-sized single-crystal 2D carbon materials with moderate bandgaps[5,9]. Here we prepare a single-crystal 2D carbon material, namely monolayer quasi-hexagonal-phase fullerene (C_{60}), with a large size via an interlayer bonding cleavage strategy. In this monolayer polymeric C_{60}, cluster cages of C_{60} are covalently bonded with each other in a plane, forming a regular topology that is distinct from that in conventional 2D materials. Monolayer polymeric C_{60} exhibits high crystallinity and good thermodynamic stability, and the electronic band structure measurement reveals a transport bandgap of about 1.6 electronvolts. Furthermore, an asymmetric lattice structure endows monolayer polymeric C_{60} with notable in-plane anisotropic properties, including anisotropic phonon modes and conductivity. This 2D carbon material with a moderate bandgap and unique topological structure offers an interesting platform for potential application in 2D electronic devices.

Two-dimensional (2D) materials are of considerable interest owing to their unique electronic and optical properties that stem from quantum confinement in the monolayer limit[1–3,10,11]. Among them, 2D carbon materials are notable. Two-dimensional carbon materials composed of conjugated carbon network structures show exceptional carrier transport characteristics that arise from their unique π-electron systems. The carbon component, as one of the most versatile elements in the periodic table, can bind to itself or almost all elements in *sp*, sp^2 and sp^3 hybridizations[12]. By altering the periodic binding motifs in hybridized carbon-atom networks, it has been shown that 2D carbon materials have great diversity in their physical properties, including tunable bandgaps that range from being metallic[4,5], semiconducting[7] or insulating[9], and they have versatile applications in transistors[4,7], energy storage devices[13] and superconductors[14]. Considerable efforts have been made towards further exploring the intriguing properties of 2D carbon networks, with these properties being closely related to the topological structures of the networks. Thus, studies investigating 2D carbon allotropes, especially those with a bandgap, have been carried out.

So far, reports on 2D materials have been limited to the structure of periodic networks interweaved by structural units of single atoms. Building 2D structures with advanced structural units, such as clusters, is a concept that has rarely been studied. Nanoclusters, known as artificial atoms, can bond in a manner similar to that of an atom to construct advanced structures[15,16]. Two-dimensional structures built by structural units of nanoclusters are expected to have superior topologies and distinctive properties; however, these structures have not yet been reported. Fullerene (C_{60}) is a typical carbon cluster. Under extremely high pressure, polymerization between C_{60} occurs by forming intercluster covalent bonds to produce a layered structure of C_{60} (ref. [17]). Such a polymeric C_{60} layer shows a regularly undulating topology of a repeating arrangement of carbon clusters in a 2D plane, with interesting electronic[18,19] and magnetic[20] properties. Nevertheless, owing to its low yield and metastability under ambient atmospheric pressure, it is quite challenging to prepare atomic-scale 2D polymeric C_{60} with a stable, ordered structure from layered polymeric C_{60}, further hindering the study of its intrinsic properties.

In this study, we prepared atomic-scale 2D polymeric C_{60}, a 2D carbon material with a framework composed of C_{60} molecules that are covalently bonded in a plane to form a periodic nanocluster network structure. Two stable magnesium (Mg)-intercalated bulk single crystals of polymeric C_{60} in closely packed quasi-hexagonal and quasi-tetragonal phases were synthesized through a facile reaction at atmospheric pressure by adjusting the ratio of Mg and C_{60}. Using an organic cation slicing strategy, monolayer polymeric C_{60} was exfoliated from the quasi-hexagonal bulk single crystals, along with few-layer polymeric C_{60} from the quasi-tetragonal bulk single crystals. Both atomic-scale 2D polymeric C_{60} materials exhibit high crystallinity and a unique topological structure. In contrast to graphene and free C_{60}, monolayer polymeric C_{60} has a moderate bandgap of approximately 1.6 eV, indicating its potential application in semiconductor devices; moreover, it demonstrates good thermodynamic stability. Furthermore, owing to the asymmetric lattice structure, significant in-plane anisotropic properties, including anisotropic phonon modes and conductivity, are observed in monolayer polymeric C_{60}, making it a promising in-plane anisotropic 2D material for functionalized electronic devices.

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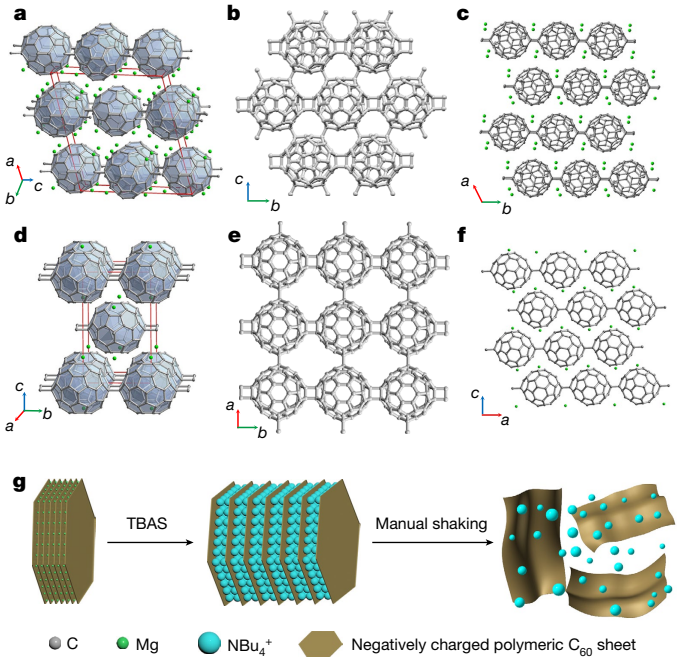


Fig. 1 | Crystal structure of qHP C_{60} and qTP C_{60} determined by single-crystal XRD. a–f, Crystal structure of the bulk single crystal of qHP C_{60} (**a**–**c**) and qTP C_{60} (**d**–**f**) from the single-crystal XRD data. **b**, **e**, The framework structure of the bulk single crystal of qHP C_{60} (**b**) and qTP C_{60} (**e**), with interlayer Mg ions hidden for a clearer illustration. C–C single bonds and [2 + 2] cycloaddition bonds are formed as bridge bonds between adjacent C_{60} cages. Each C_{60} cage forms bridge bonds with six adjacent cages in qHP C_{60} and with four adjacent cages in qTP C_{60}. **c**, **f**, Side view of the bulk single crystal of qHP C_{60} (**c**) and qTP C_{60} (**f**). Notably, *a*–*b* stacking of the polymeric C_{60} layers is displayed in the single-crystal qTP C_{60}. By stacking the polymeric C_{60} layer, pseudotetrahedral and pseudooctahedral holes are formed in the bulk single crystal of qHP and qTP C_{60}. In single-crystal qHP C_{60}, the Mg ions occupy both the pseudotetrahedral and pseudooctahedral holes. In single-crystal qTP C_{60}, the Mg ions occupy only the pseudotetrahedral holes. **g**, Schematic of organic cation slicing exfoliation. The interlayer Mg ions (green spheres) are replaced by larger NBu_4^+ ions (blue spheres) after the interlayer bonding between polymeric C_{60} layers is cleaved when the Mg ions coordinate with salicylate. Nanosheets of negatively charged polymeric C_{60} were obtained through gentle manual shaking.

Fabrication and characterization

Atomic-scale polymeric C_{60} was fabricated from bulk single crystals through an organic cation slicing strategy. C_{60} molecules were first polymerized through a facile reaction with the intercalation of Mg to form Mg-intercalated bulk single crystals of polymeric C_{60} (Methods). The single-crystal X-ray diffraction (XRD) results show that the Mg-intercalated bulk single crystals have a layered structure of alternating polymeric C_{60} layers and Mg atomic layers (Fig. 1). The polymeric C_{60} layer consists of a fluctuating 2D network of spherical C_{60} nanoclusters in which neighbouring C_{60} cages are interconnected through bridge bonds, that is, C–C single bonds or [2 + 2] cycloaddition bonds, and the same bridge bonds between C_{60} cages are linearly arranged in the same direction. Two configurations of Mg-intercalated bulk single crystals, namely, a closely packed quasi-hexagonal phase (qHP) and a quasi-tetragonal phase (qTP), are obtained by varying the ratio of C_{60} and Mg during polymerization (Fig. 1a–f). According to the single-crystal data, the compositions in the qHP and qTP single crystals are Mg_4C_{60} and Mg_2C_{60} (ref. [21]), respectively. In the qHP bulk single crystals, each C_{60} cage is linked with six neighbouring C_{60} cages by bridge bonds separated by approximately 60°, including the [2 + 2] cycloaddition of '5, 6' bonds along the [010] direction and C–C single bonds along both the [011] direction and [01$\bar{1}$] direction. Regarding the qTP bulk crystals, one C_{60} is connected to four adjacent C_{60} cages through bridge bonds spaced approximately 90° apart, wherein the [2 + 2] cycloaddition of '6, 6' bonds occurs along the [010] direction with the C–C single bonds forming along the [100] direction. It is noted that the quantity of the bridge bonds on one C_{60} cage can affect the distribution of π states in the cage. After polymerization, there are only 52 and 54 π-state carbon atoms in each C_{60} for the qHP and qTP bulk single crystals, respectively (Extended Data Fig. 1). Thus, electrical properties that are distinct from those in free C_{60} molecules are potentially generated in polymeric C_{60}. In addition, the single-crystal XRD data show that the C_{60} cages in the polymeric C_{60} layer stretch along the direction of the [2 + 2] cycloaddition bond (Fig. 1a, d). The orderly arrangement of these asymmetric C_{60} cages leads to an asymmetric 2D layer, which is expected to lead to the anisotropic physical properties of 2D polymeric C_{60}.

In the Mg-intercalated bulk single crystal polymeric C_{60}, adjacent C_{60} layers are held together by Mg–C linking bonds. The lengths of the C–Mg bonds measured from the single-crystal XRD data are 2.22–2.37 Å; thus, these bonds are between being covalent and being electrovalent. Differing from the weak interlayer van der Waals force in traditional layered 2D materials, the stronger interlayer Mg–C bonds make the bulk polymeric C_{60} crystal unable to be exfoliated by mechanical exfoliation. In addition, these bulk crystals cannot be cleaved in polar solvents, as in the case of alkali-metal-intercalated 2D materials, because the C–Mg bonds are difficult to ionize[22]. Thus, an organic cation slicing strategy is proposed to effectively exfoliate bulk single crystals into atomic-scale flakes (Fig. 1g). The Mg ions, which are alkaline earth metal ions, more easily coordinate with oxygen-containing salicylate. Therefore, we chose tetrabutylammonium salicylate (TBAS) to react with the bulk single crystals. After the coordination of Mg ions with salicylate, the interlayer C–Mg bonds between polymeric C_{60} layers are cleaved, and then tetrabutylammonium (NBu_4^+) cations intercalate into the interlayer of the polymeric C_{60} crystals, replacing the Mg ions to maintain the charge balance (the generated magnesium salicylate is verified by mass spectrometry, as shown in Extended Data Fig. 2a). The NBu_4^+ cations enable an effective expansion of the interlayer space, owing to the larger ionic radius than Mg. Expansion of the interlayer space is demonstrated by the movement of the (200) peak in the XRD spectra (Extended Data Fig. 2b). Coupled with the good solubility of NBu_4^+, the NBu_4^+-intercalated polymeric C_{60} crystals are readily exfoliated and dispersed, with only gentle manual shaking and no sonication (as detailed in Methods). The larger interlayer space and higher interlayer Mg-ion reactivity in the qHP bulk crystals of C_{60} are favourable for exfoliating monolayers with cation slicing. Two kinds of polymeric C_{60} flakes, the monolayer qHP C_{60} and few-layer qTP C_{60}, were obtained from their bulk crystals through organic cation slicing. The obtained flakes are negatively charged and countered by NBu_4^+ cations, where the monolayer qHP C_{60} can be described as $(NBu_4^+)_6C_{60}^{6-}$ (as detailed in Methods).

The two atomic-scale polymeric flakes were first identified under an optical microscope, with lateral dimensions of approximately 60 μm and 30 μm, respectively (Fig. 2a, e). The precise flake thickness was obtained by atomic force microscopy (AFM). The qHP C_{60} flake is measured to have a thickness of 1.22 nm, and the qTP C_{60} flake has a thickness of 2.63 nm (Fig. 2b, f). As the diameter of the C_{60} cage is 0.7 nm, the obtained qHP C_{60} flake is supposed to be monolayer and the qTP C_{60} flake is few-layer, consistent with the optical observations. Furthermore, the sizes and yield of monolayer qHP C_{60} were investigated by statistical analysis, as shown in Extended Data Fig. 3. The exfoliated monolayer qHP C_{60} flakes mostly range from 30 μm to 80 μm, with a yield of approximately 90%.

We carefully studied the crystal structure of monolayer qHP C_{60} and few-layer qTP C_{60} with transmission electron microscopy (TEM) and aberration-corrected scanning transmission electron microscopy (STEM). The thin flakes of qHP C_{60} and qTP C_{60} are evident by the low contrast in the TEM image (Fig. 2c, g). The selected area electron diffraction (SAED) measurements show obvious diffraction patterns in both

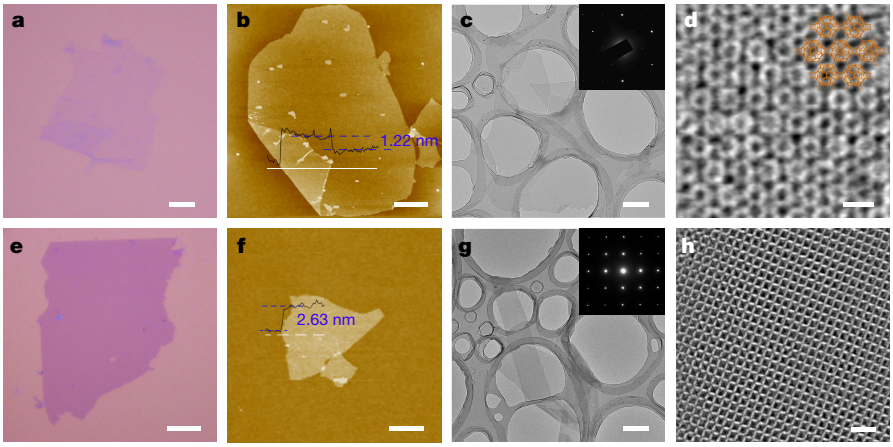


Fig. 2 | Monolayer qHP C_{60} and few-layer qTP C_{60}. **a**, **e**, Optical images of monolayer qHP C_{60} (**a**) and few-layer qTP C_{60} (**e**). **b**, **f**, AFM images of monolayer qHP C_{60} (**b**) and few-layer qTP C_{60} (**f**). **c**, **g**, TEM characterization of monolayer qHP C_{60} (**c**) and few-layer qTP C_{60} (**g**). SAED patterns of the corresponding flakes are shown in the insets. **d**, **h**, STEM images of monolayer qHP C_{60} (**d**) overlaid with chemical structures and few-layer qTP C_{60} (**h**). Scale bars, 10 μm in **a**; 1 μm in **b**, **c**, **g**; 5 μm in **e**; 2 μm in **f**; 1 nm in **d**; and 2 nm in **h**.

flakes, suggesting the high crystallinity of both atomic-scale polymeric C_{60} structures. The SAED patterns (insets in Fig. 2c, g) of the qHP C_{60} and qTP C_{60} flakes are consistent with the characteristic lattices in the corresponding polymeric C_{60} crystal, as obtained from the single-crystal data. Atomically resolved STEM images were collected at a low accelerating voltage of 60 kV and with ångström-scale electron beam scans over the samples. The quasi-hexagonal and quasi-tetragonal lattices are clearly visible in the respective STEM images of monolayer qHP C_{60} and few-layer qTP C_{60} (Fig. 2d, h). Owing to the monolayer structure, the unique 2D topology structure composed of C_{60} nanocluster units is clearly shown in the STEM image of monolayer qHP C_{60}, where individual spherical C_{60} cages are distinguishable as a 'circle' with identifiable bridge bonds between them (Fig. 2d). The distance between adjacent C_{60} centres is approximately 9.1 Å from the STEM image, which agrees with that of the bulk crystal of qHP C_{60}. Regarding few-layer qTP C_{60}, a periodic square lattice is seen in the STEM image, corresponding to the *a*–*b* stack of the qTP C_{60} layers along the *c* axis, as shown in the single-crystal data (Fig. 2h, Supplementary Information).

Owing to the characteristics of a single layer, we concentrated on the related research of monolayer qHP C_{60}. Figure 3a shows the Raman spectra of qHP C_{60} bulk single crystals and monolayers. Obvious vibrational peaks are observed at 1,403 cm^{-1} and 1,409 cm^{-1} for the qHP C_{60} bulk single crystal and the monolayer, respectively, corresponding to the characteristic vibration peak A_g^2 of C_{60}. The A_g^2 mode arising from the pentagonal-pinch mode has been used universally as an analytical probe for the structural and electronic properties of C_{60}. Compared with 1,469 cm^{-1} in pristine C_{60}, the A_g^2 mode downshifts significantly in the qHP C_{60} bulk single crystals and monolayers. This behaviour is attributed to the formation of intermolecular bridge bonds changing the geometry of individual C_{60} (ref. [23]), and the addition of electrons to C_{60} softens the bond stretching modes as the added electrons enter antibonding molecular orbitals[24]. Two moderate peaks at 959 cm^{-1} and 985 cm^{-1} are observed in the qHP C_{60} bulk single crystals, which broaden into two respective wide peaks at 907 cm^{-1} and 962 cm^{-1} in the monolayer. The two peaks can be attributed to the bridge bonds and correspond to the stretch mode of the four-carbon ring[25,26]. In addition, owing to the reduced symmetry in the qHP C_{60} bulk single crystal and monolayer, some previously forbidden modes of pristine C_{60} reappear in the Raman spectra.

Next, we investigated the electronic structure of monolayer qHP C_{60} with ultraviolet photoemission spectroscopy (UPS) and low-energy inverse photoemission spectroscopy (LEIPS). As shown in Fig. 3b, the valence-band maximum (VBM) of monolayer qHP C_{60} is located at about 1.17 eV below the Fermi level (E_F), and the conduction-band minimum (CBM) is positioned at about 0.43 eV above the Fermi level. Consequently, monolayer qHP C_{60} is determined to be a semiconductor with a transport bandgap of about 1.6 eV; this result indicates the transformation of C_{60} from an insulator to a semiconductor. The optical bandgap is also measured as about 1.55 eV, as shown in Extended Data Fig. 4. As monolayer qHP C_{60} is a semiconductor, it is desirable to explore its field-effect transistor properties. We fabricated ionic liquid-gated field-effect transistors on monolayer qHP C_{60}, which exhibit typical n-type transport behaviour. The devices show an electronic mobility of 1–5 $cm^2 V^{-1} s^{-1}$, with an on/off ratio of over 20 and a low threshold voltage of 0.5 V (Extended Data Fig. 5).

Measurement of anisotropy

The asymmetric alignment of bridge bonds leads to two inequivalent directions within the in-plane lattice of monolayer qHP C_{60} (Fig. 1b). Therefore, we performed angle-resolved polarized Raman spectroscopy (ARPRS) to study the phonon modes associated with lattice symmetry. A clear correlation with the polarized angles is observed from the A_g^2 mode at 1,409 cm^{-1}. The angle-dependent intensities of the A_g^2 mode were extracted to plot in a polar figure (Fig. 4a). On the basis of the classical Raman selection rules, the polarization-dependent Raman

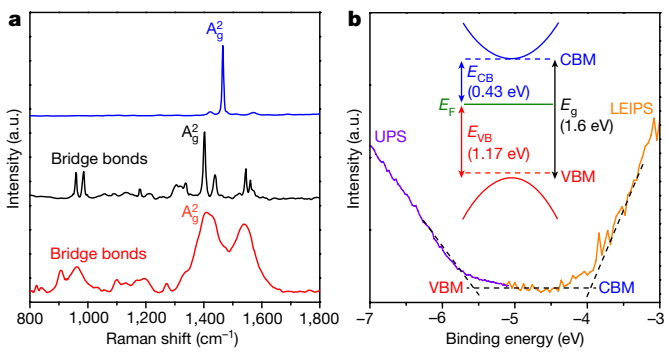


Fig. 3 | Raman spectra and band structure of qHP C_{60}. **a**, Raman spectra of intrinsic C_{60} (blue line), qHP C_{60} bulk crystal (black line) and exfoliated monolayer qHP C_{60} nanosheets (red line). **b**, Electronic band structure of the monolayer qHP C_{60} nanosheets measured by the UPS and LEIPS experiments. E_g represents the gap energy; E_{CB} represents the energy difference between CBM and E_F; and E_{VB} represents the energy difference between VBM and E_F.

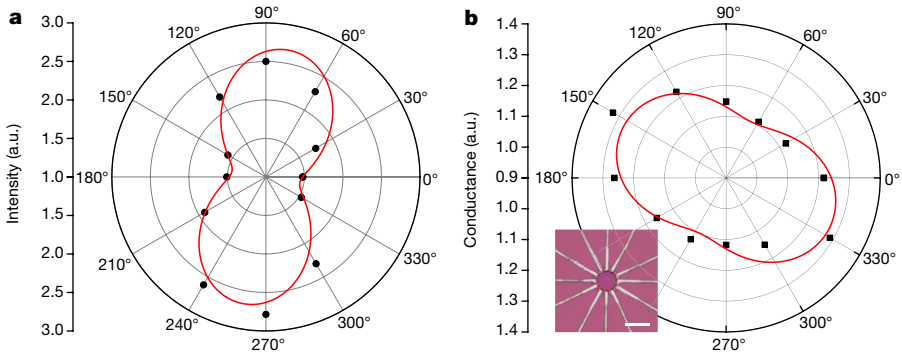


Fig. 4 | Anisotropic properties of qHP C_{60}. a, Angle-resolved Raman spectrum of monolayer qHP C_{60}. Raman intensities of the A_g^2 mode (black dots) are collected with incident light at different polarization angles. The intensity of the A_g^2 mode as a function of the excitation laser polarization angle was fitted according to classical Raman selection rules, and the fitting result is represented by the solid red line. **b**, Pole figure of the conductance of a monolayer qHP C_{60} device along 12 directions that are evenly spaced apart at an angle of 30° (black dots). The solid red line represents the fitting curve of device conductance as a function of the direction angle. The monolayer qHP C_{60} device is shown in the bottom left of **b**. The Raman anisotropy and electrical conductivity anisotropy are measured with the same sample. Scale bar, 5 µm.

intensity of the A_g^2 mode (I) can be described as $I = (A\cos^2(\theta+\varphi) + C\sin^2(\theta+\varphi))^2$, where A and C are constants of the Raman tensor, θ is the angle of incident polarization relative to the 0° reference, and φ corresponds to the angle between the b axis and the 0° reference. The fitted curve agrees well with the experimental data. The A_g^2 mode exhibits a clear periodic dependence on the polarization angles and has two-fold symmetry in the polar figure, with a maximum at 80° and a minimum at 170°. The ARPRS measurement clearly reveals the vibrational anisotropy in monolayer qHP C_{60}.

As anisotropic electron–phonon interactions greatly affect the electrical properties, we examined the electron transport in monolayer qHP C_{60} by angle-resolved conductance measurements. As shown in inset of Fig. 4b, current–voltage measurements were taken along each pair of electrodes. The results show a periodic dependence on the testing angle, indicating the in-plane anisotropic conductivity of the sample. According to the conductivity equation that includes the direction angle of anisotropic materials (Methods), the ratio of conductance along the maximum direction and minimum direction is calculated to be approximately 1.2. Furthermore, by comparing the Raman anisotropy with the electrical conductivity anisotropy of the same sample, it is found that the direction of the minimum Raman intensity of A_g^2 corresponds to a maximum conductivity.

Finally, the 2D carbon material qHP C_{60} shows excellent environmental stability and thermal stability (Extended Data Fig. 6). We also find out that hydrogen peroxide (H_2O_2) can quench the negative charges without further oxidization, and remove the NBu_4^+ on the polymeric C_{60} flakes, all of which will produce a neutral carbon material (discussed in Methods).

In conclusion, we have successfully fabricated stable crystalline monolayer polymeric C_{60} with a cluster-based topology in a periodic arrangement. The asymmetric lattice structure of monolayer qHP C_{60} led to obvious in-plane anisotropic properties, such as anisotropic conductivity in two-fold symmetry. The moderate bandgap of approximately 1.6 eV of monolayer polymeric C_{60} bridges the energy gap between semimetallic graphene and large-bandgap fullerene, making polymeric C_{60} a potential candidate for use in 2D optoelectronic devices. Our demonstration of monolayer polymeric C_{60} is expected to inspire a fresh perspective in the exploration of 2D carbons with unique topologies and properties.

Online content

Any methods, additional references, Nature Research reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-022-04771-5.

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Methods

Synthesis of polymeric C_{60} single crystals

C_{60} powder (Sigma-Aldrich, 99.5+%) was first purified by sublimation. A mixture of the purified C_{60} and Mg metal powder (Sigma-Aldrich, 300 mesh) was vacuum-sealed in a quartz tube, which was subsequently placed in a two-zone furnace with a temperature of 500–580 °C for 20 h. The mixture of Mg and C_{60} powder was put in the higher-temperature zone, whereas the polymeric C_{60} single crystals were grown in the lower-temperature zone. As Mg can react with quartz tube at high temperature, the control of the crystal-growing temperature and molar ratio of Mg:C_{60} are crucial in growing the single crystals. In our case, single crystals of qHP C_{60} (CCDC 2159850) were synthesized at a Mg:C_{60} molar ratio of 15:1 and the single crystals of qTP C_{60} (CCDC 2159902) were synthesized at a Mg:C_{60} molar ratio of 9:1.

Exfoliation of bulk single crystal of qHP C_{60} and qTP C_{60}

TBAS was dissolved in *N*-methylpyrrolidone (NMP) to obtain a saturated solution. The polymeric C_{60} single crystals were added to TBAS solution under inert atmosphere for over 7 days. After manual shaking, the mixture was centrifuged at 1,000 rpm for 10 min to remove the sediment, and then centrifuged at 8,000 rpm for 10 min to remove TBAS solution. Exfoliated polymeric C_{60} was washed up with NMP and finally dispersed into NMP for other applications.

Characterization of negative charges on monolayer qHP C_{60}

XPS and zeta-potential characterization were utilized to clarify the nature of the negative charge on monolayer flakes. From the XPS investigations of qHP C_{60} before in Extended Data Fig. 7, the negligible atomic ratio of Mg in the spectrum indicates the absence of Mg in monolayer qHP C_{60}, and the retention of TBA cations on monolayer qHP C_{60} is clearly identified. Therefore, Mg ions were completely replaced by NBu_4^+ cations. Moreover, considering the ratio of the C–N peak and the C–C peak in the C 1*s* XPS spectra, the proportion of C_{60}:NBu_4^+ is 1:6 in monolayer qHP C_{60} (Extended Data Fig. 8). Besides, the zeta potential of monolayer qHP C_{60} solution (Extended Data Fig. 9) is measured as −40 mV, indicating the existence of negative charge on monolayer flakes. Meanwhile, negative charge on monolayer flakes can be neutralized by treatment with H_2O_2, with NBu_4^+ cations being eliminated. H_2O_2 quenched only the negative charges on polymeric C_{60} flakes and did not further oxidize the C_{60} cages, as the typical C–O peak at 286–287 eV was absent (Extended Data Fig. 10).

Measurement of UPS and LEIPS

The UPS measurements were performed on an Axis Ultra DLD (Kratos) spectrometer with an unfiltered He I (21.22 eV) excitation source and a pass energy of 5 eV. A bias voltage of −9 V was applied to the sample to obtain the cut-off region of secondary electrons. The LEIPS measurements were performed on a PHI5000 VersaProbe III instrument (ULVAC PHI) in Bremsstrahlung isochromatic mode.

ARPRS characterization

Raman spectra were excited by a 532-nm helium–neon laser, and the linear polarizer was placed parallel to the incident photon polarization. The incident light is polarized along the horizontal direction, with rotation angles in steps of 30°. A parallel polarization configuration is employed in the measurement, with only the Raman signals polarized parallel to the incident light being collected. Raman shifts were set ranging between 1,300 cm^{-1} and 1,700 cm^{-1} to investigate the characteristic A_g^2 mode located at around 1,409 cm^{-1}.

The intensity of Raman vibration modes can be described as $I \propto |\mathbf{e_i} \bullet R \bullet \mathbf{e_s}|^2$, where $\mathbf{e_i}$ is the unit vector of incident laser polarization, $\mathbf{e_s}$ is the scattering phonon polarization and R is the Raman tensor. In our case, $\mathbf{e_i}$ is parallel to $\mathbf{e_s}$. By defining the *b* axis and *a* axis directions as *x* and *z*, respectively, thus $\mathbf{e_i} = \mathbf{e_s}^T = (\cos\omega, 0, \sin\omega)$, where ω is the angle between incident polarization and the *b* axis. For the A_g^2 mode with C_{2h} symmetry, the Raman tensor R can be described as

$$R = \begin{pmatrix} A & D & 0 \\ D & B & 0 \\ 0 & 0 & C \end{pmatrix}$$

where A, B, C and D are constants of the Raman tensor. Therefore, the intensity of A_g^2 mode can be written as $I \propto (A\cos^2\omega + C\sin^2\omega)^2$. By setting $\omega = \theta + \varphi$, where θ is the angle of incident polarization relative to the 0° reference, and φ corresponds to the angle between the *b* axis and the 0° reference, the fitting curve of Raman intensity can be described as $I = (A\cos^2(\theta + \varphi) + C\sin^2(\theta + \varphi))^2$.

Electronic conductivity anisotropy characterization

Freshly exfoliated qHP C_{60} nanosheets were transferred onto a silicon dioxide substrate and baked at 130 °C for 5 min. Then, the patterns of 12 terminal qHP C_{60} devices were defined using electron-beam lithography. Chromium/gold (10 nm/40 nm) was deposited as contact electrodes using electron-beam evaporation. Six pairs of electrodes (30° between each two electrodes) were measured to characterize the d.c. conductance. The distance between each pair of electrodes was 5 μm.

The conductivity of the anisotropic semiconductor can be described by the equation $\sigma_\theta = \sigma_{min}\sin^2(\theta - \varphi) + \sigma_{max}\cos^2(\theta - \varphi)$ (refs. [27,28]), where σ_{min} and σ_{max} refer to the conductivity along the maximum and minimum directions, respectively. θ is the angle of current flow with respect to the 0° reference direction. σ_θ is the conductivity along the current flow direction. φ is the angle between the σ_{max} direction and the 0° reference. In our case, $\varphi = -23°$ gives the best fit for this sample. A sample of the angle-resolved d.c. conductance measurements was subsequently measured by ARPRS at room temperature.

Characterizations

Characterizations were carried out using TEM (JEM-2100F; acceleration voltage, 200 kV) with SAED, STEM (Spectra 300; acceleration voltage, 60 kV; detector, Panther STEM-ABF), Raman (Horiba Labram HR Evolution; 532-nm laser wavelength), ultraviolet–visible spectroscopy (Shimadzu UV-2600), XPS (ESCALab250Xi), single-crystal XRD (Synergy-R), powder XRD (Empyrean), AFM (Bruker Dimension Icon Scanning Probe Microscope) and zeta potential (Malvern Zetasizer Nano ZSE).

Data availability

All data supporting the findings of this work are available within the paper and its Extended Data. Source data are provided with this paper.

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28. Xia, F., Wang, H. & Jia, Y. Rediscovering black phosphorus as an anisotropic layered material for optoelectronics and electronics. *Nat. Commun.* **5**, 4458 (2014).

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Author contributions J.Z. designed the research. L.H. performed the exfoliation approach and analysed the results. X.C., S.W., R.L., Y.L. and D.Z. assisted with the exfoliation approach and the analysis of the results. B.G. performed the electron microscopy analyses. J.Z. supervised the research. J.Z., L.H. and X.C. co-wrote the paper. All authors discussed the results and commented on the manuscript.

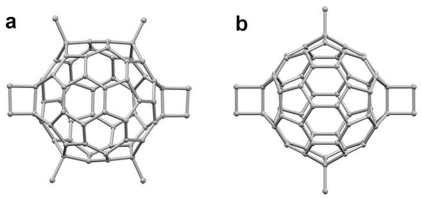
Competing interests The authors declare no competing interests.

Additional information
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41586-022-04771-5.

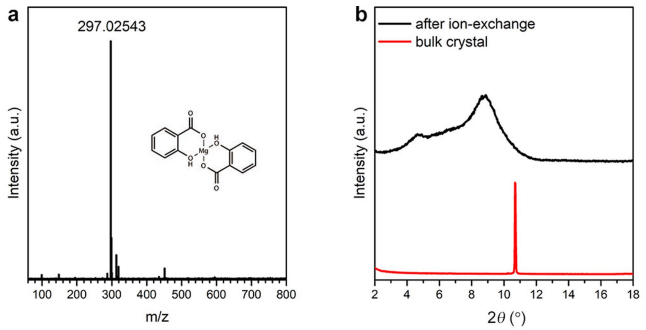
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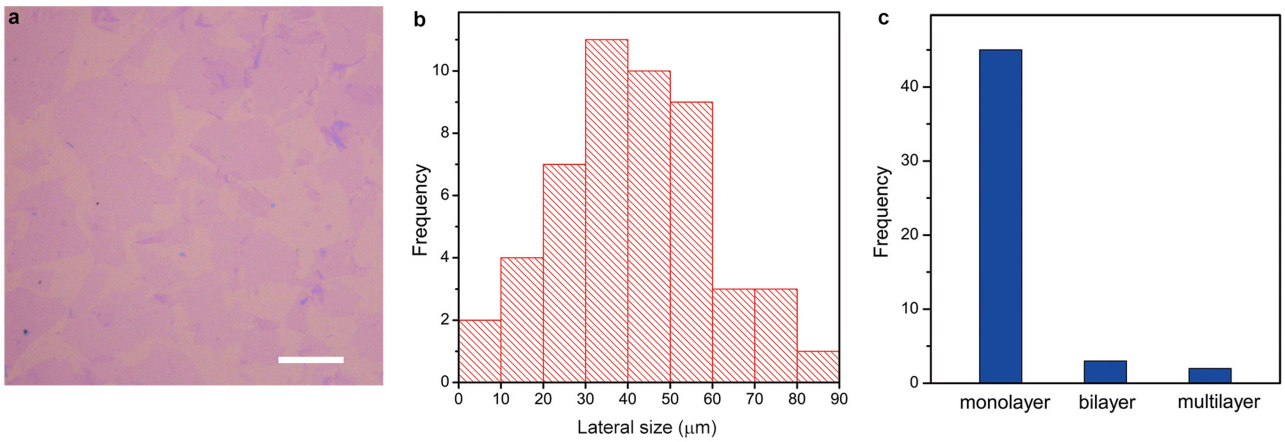
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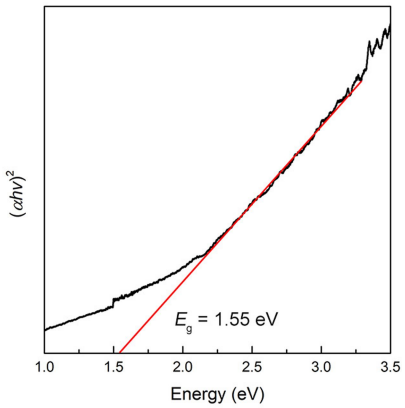
Extended Data Fig. 1 | Schematic illustration of C_{60} cage in qHP C_{60} and qTP C_{60}. a, In qHP C_{60}, each C_{60} cage forms two [2+2] cycloadditon C-C bonds and four C-C single bonds with neighbouring C_{60} cages. Thus there are only 52 carbon atoms possess π states in each C_{60} cage. **b**, Differing from that in qHP C_{60}, two [2+2] cycloadditon C-C bonds and two C-C single bonds are formed with neighbouring C_{60} cages in qTP C_{60}, therefore 54 carbon atoms possess π states in each C_{60} cage.



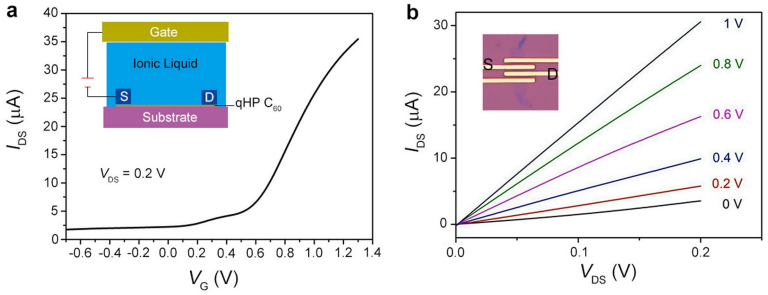
Extended Data Fig. 2 | Exfoliation mechanism exploration. a, Magnesium salicylate is obtained from the reaction system by chromatographic separation. The peak at 297.02543 corresponds to $[M\text{-}H]^-$ signal of magnesium salicylate. **b**, XRD spectra of Mg intercalated qHP C_{60} before (red) and after (black) ion-exchange reaction. In Mg intercalated bulk qHP C_{60}, interlayer distance is characterized by (200) peak in $2\theta = 10.7°$. After ion-exchange reaction, the (200) peak moves to lower 2θ position, which indicate the expanding of interlayer distance. The peak in $2\theta = 8.8°$ (interlayer distance of 10 Å) is attributed to the intercalation of NBu_4^+ ions. As NBu_4^+ ions are flexible, the expanded (200) peaks in exfoliated polymeric C_{60} are broader than that in bulk crystals.



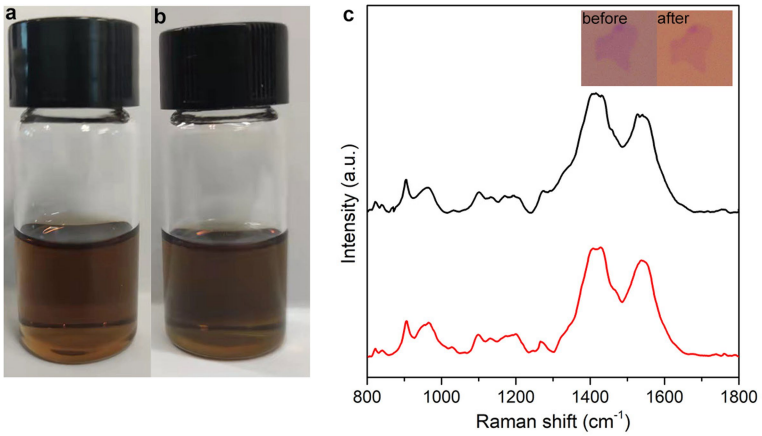
Extended Data Fig. 3 | Statistical analysis of the lateral sizes and proportion of monolayer for exfoliated qHP C_{60} flakes. a, Typical optical image of the exfoliated qHP C_{60} flakes on SiO_2/Si substrate. Scale bar, 50 μm. **b**, The statistics of lateral sizes of monolayer qHP C_{60}. **c**, Statistics of thickness of exfoliated qHP C_{60} flakes.



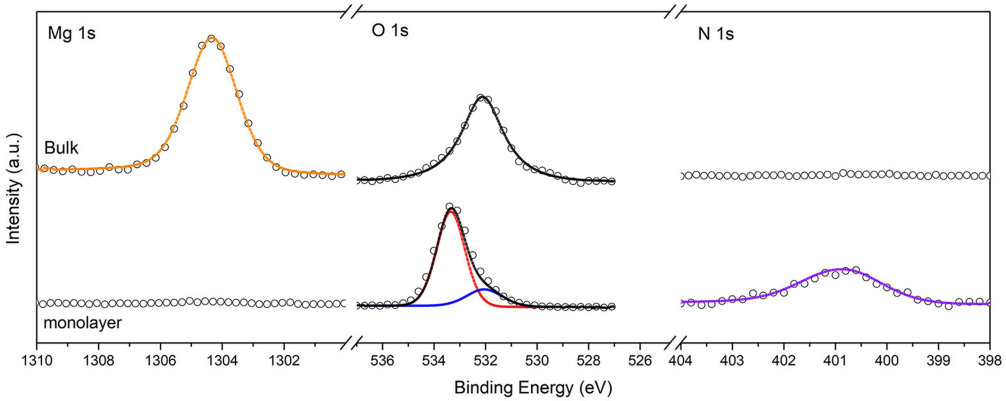
Extended Data Fig. 4 | Optical bandgap of exfoliated qHP C_{60}. Tauc plot to determine the optical bandgap of qHP C_{60}, with extrapolation of the linear region (red line) estimating of approximately 1.55 eV.



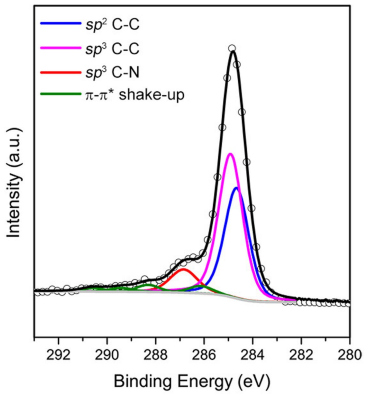
Extended Data Fig. 5 | Monolayer qHP C_{60} transistors. a, Transfer characteristics of monolayer qHP C_{60} transistors. Inset is a schematic of the device. **b**, Output characteristics of the device in **a**.



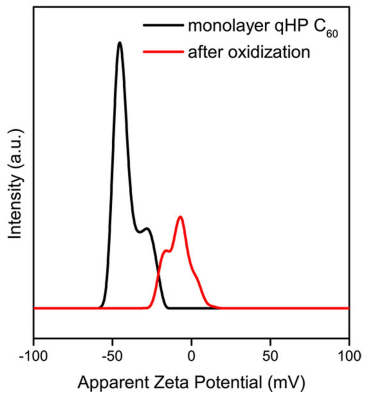
Extended Data Fig. 6 | Environment stability (a, b) and thermal stability (c) of the exfoliated qHP C_{60}. a, **b**, Photograph of exfoliated qHP C_{60} nanosheets dispersed in NMP before (**a**) and after (**b**) standing in air for one month. The solution retained clear without sediment or aggregation. **c**, Raman spectra and optical images of exfoliated qHP C_{60} flakes before (red) and after (black) heating at 600 K for ten minutes. The Raman spectrum and optical image remain unchanged after heating, indicating that the polymeric C_{60} frameworks in exfoliated qHP C_{60} do not decompose at 600 K.



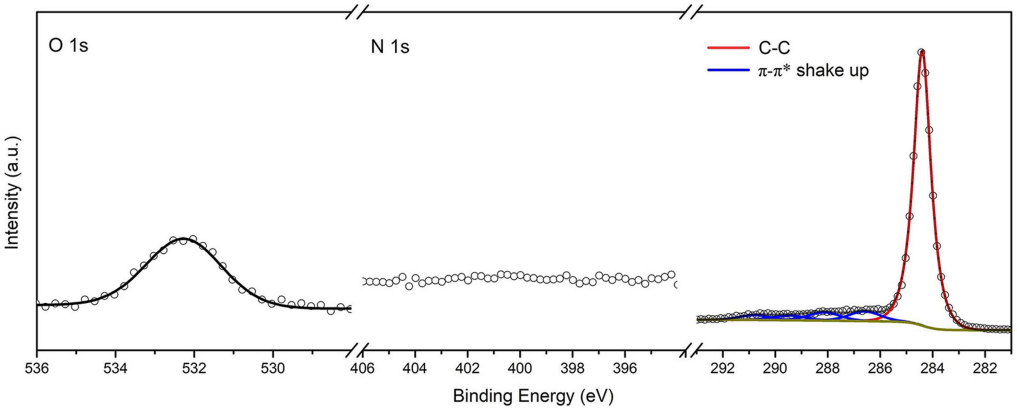
Extended Data Fig. 7 | XPS spectra for bulk qHP C_{60} (upper) and monolayer qHP C_{60} (bottom). For monolayer qHP C_{60}, peak of Mg 1s is absent, and the appearing N 1s peak at 401 eV binding energy is typical for quaternary ammonium groups. The O 1s (532 eV) and O 1s (533 eV) peaks correspond to oxygen from surrounding environment and the SiO_2 substrate, respectively. Meanwhile, absence of typical carboxyl group peak in O 1s (535 eV) indicates that there is no residual salicylate on monolayer flakes.



Extended Data Fig. 8 | C 1s XPS spectra for monolayer qHP C_{60}. C 1s peak can be consistently deconvoluted into several components attributed to sp^2-hybridized C-C (284.6 eV), sp^3-hybridized C-C (284.9 eV), sp^3-hybridized C-N (286.8 eV) with approximately 52:80:24 intensity ratios, in agreement with the C_{60}/NBu_4^+ ratio of 1:6 in monolayer qHP flakes. Moreover, a series of π-π* shake-up satellite peaks of C_{60} are identified in 286.2eV, 288.3 eV, 289.4 eV and 290.6 eV.



Extended Data Fig. 9 | Zeta potential distribution of monolayer qHP C_{60} solution before (black) and after (red) treatment with hydrogen peroxide. Average zeta potential of monolayer qHP C_{60} solution is calculated as −40.2 mV, indicating the negative charge on monolayer flakes. After oxidization, negative charges are neutralized, and average zeta potential turns to −8.4 mV.



Extended Data Fig. 10 | XPS spectra for 3% H_2O_2 treated monolayer qHP C_{60}. The absence of N 1s peak indicates the removal of NBu_4^+ cations on the monolayer after the treatment of H_2O_2. Besides, C-O bonds are not formed, because typical C-O peak at 286–287 eV is not found. The O 1s peak is attributed to the adsorbed oxygen from the surrounding environment.